Representation Learning

DS677001 Fall 2025

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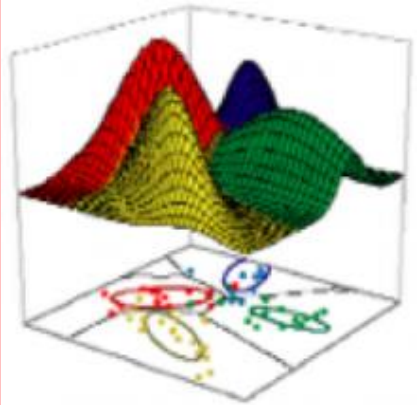
A Rudimentary Machine Learning Pipeline

Extract
Representations

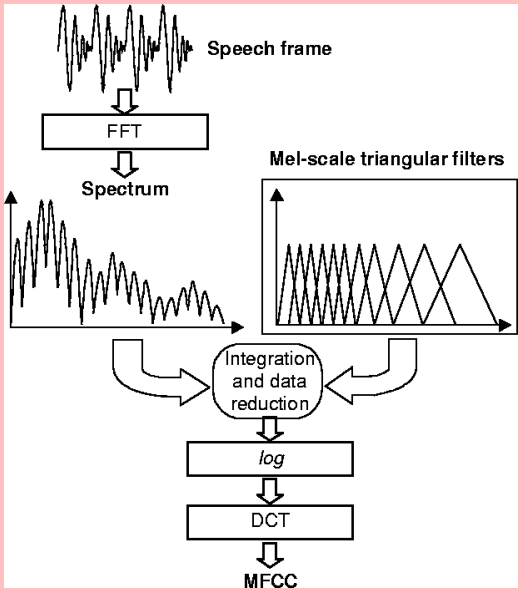


Build
Classifier/Regression
model

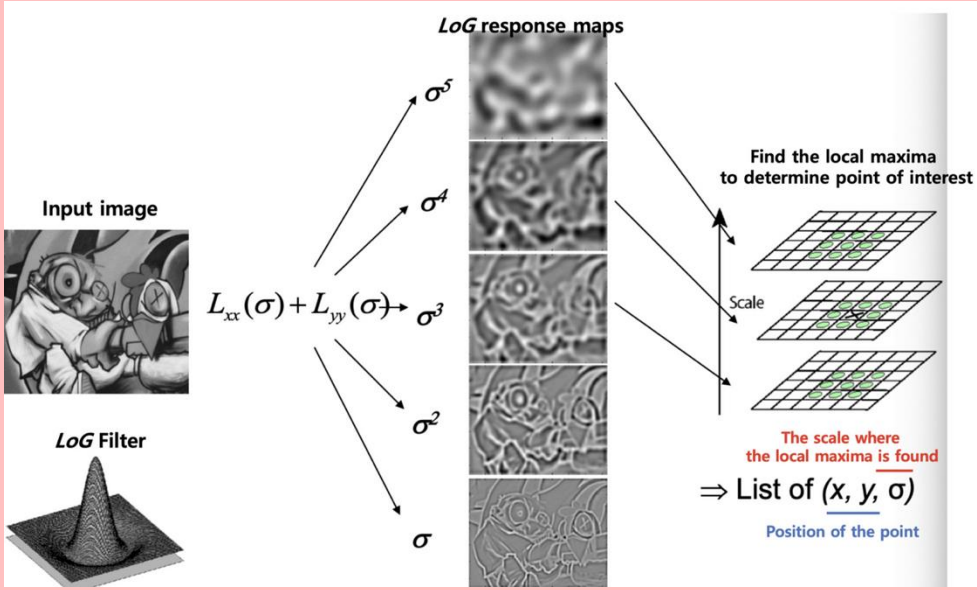
Representations aid learning

- Consider arithmetic with Roman numerals vs Arabic numerals
- $CCCXXI \times IV = MCCLXXXIV$
- $321 \times 4 = 1284$
- A good representation makes "explicit certain entities or types of information" (Marr 1982)

Feature Engineering



Mel Frequency Cepstral Coefficients - Speech



Scale Invariant Feature Transform - Images

$$w_{i,j} = tf_{i,j} \times \log\left(\frac{N}{df_i}\right)$$

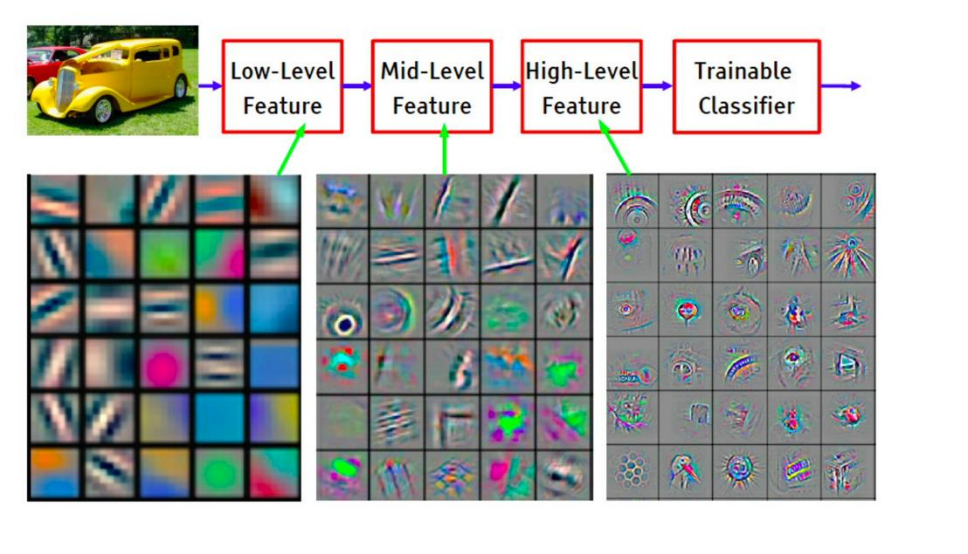
tf_{ij} = number of occurrences of i in j
df_i = number of documents containing i
N = total number of documents

TF-IDF for documents

Deep Networks learn representations

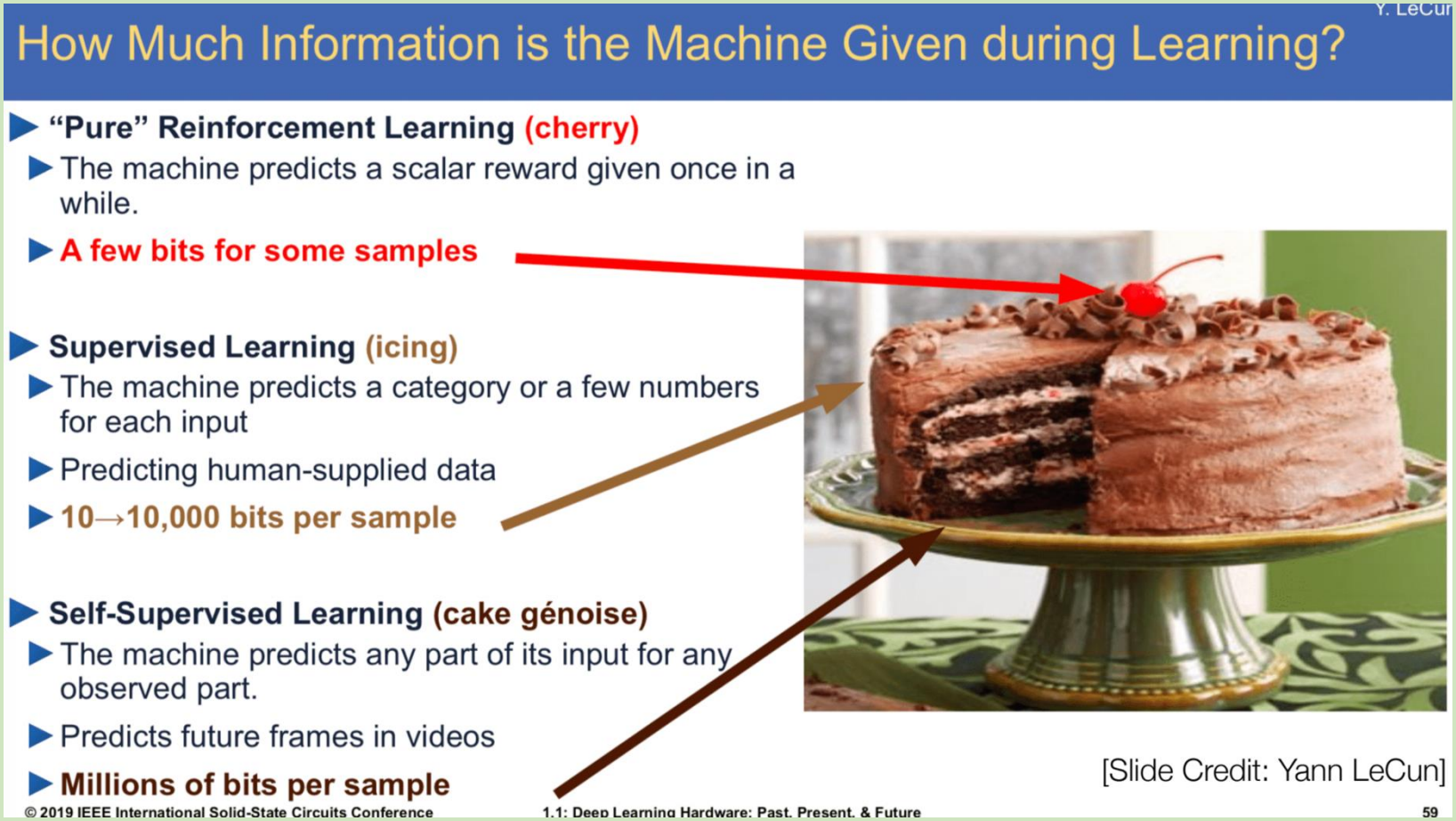
- Instead of hand-crafting representations, deep networks learn representations

- Each layer is a different representation of the data



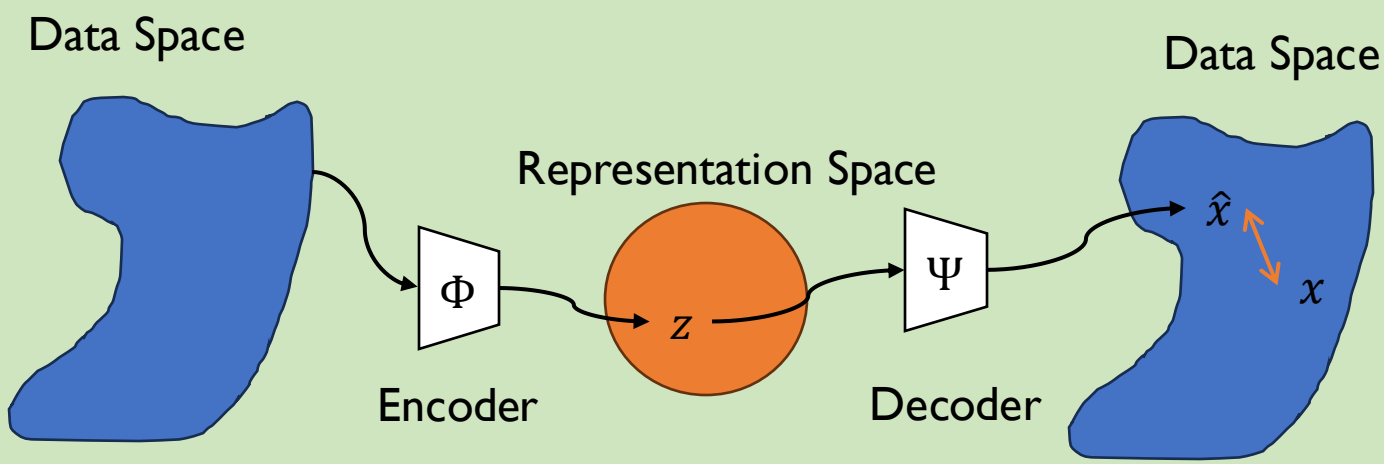
Slide courtesy Y. Lecun

Self-supervised learning is (the most?) important



Reconstruction based Representation learning

$$\min_{\Psi,\Phi} \frac{1}{n} \sum_{i=1}^{n} \left|\left| x_i - \Psi \circ \Phi(x_i) \right|\right|^2$$



Learning Representations – Desiderata

- Compact(minimal)
- Explanatory(sufficient)
- Disentangled(independent factors)
- Interpretable
- Make subsequent problem solving easy

- For classification: concentration, separation, robustness to irrelevant perturbations

Linear Representation Learning

Principal Components Analysis

$$\min_{\Psi,\Phi} \frac{1}{n} \sum_{i=1}^{n} \left|\left| x_i - \Psi \circ \Phi(x_i) \right|\right|^2$$

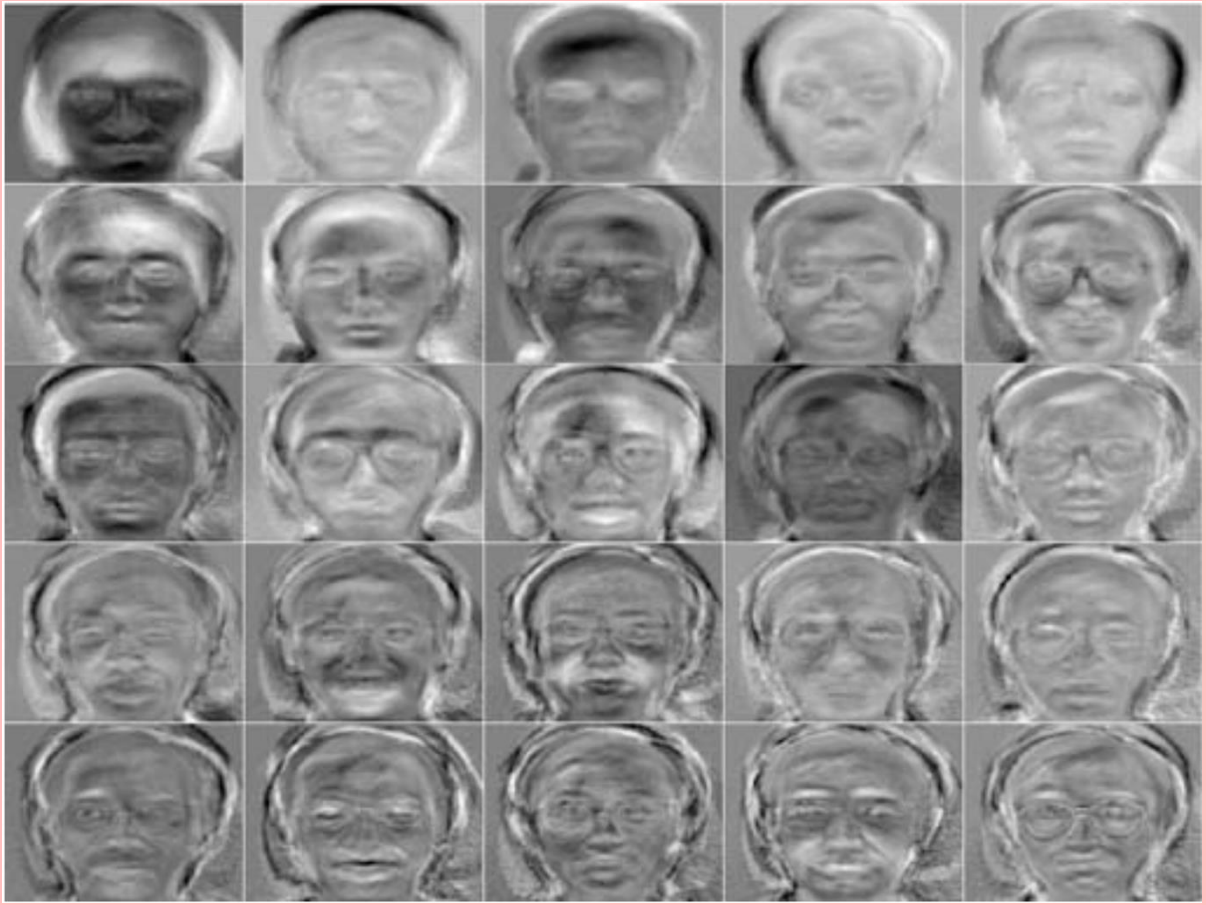
$\Psi = U \ \in \mathbb{R}^{d \times k}, k \leq d,$ orthogonal columns
$\Phi = U^T$

- Statistical view of PCA
- Geometric view of PCA

Example – Eigenfaces [Turk and Pentland, 1991]

- Face recognition problem
- $n = 500$ images of size $d = 256 \times 256$
- Find the principal subspace of dimension $k = 25$
- Map new images to the principal components – perform classification in this space

Eigenfaces [Turk and Pentland, 1991]



Dictionary Learning or Sparse Coding

$$\min_{\Psi,\Phi} \frac{1}{n} \sum_{i=1}^{n} \left|\left| x_i - \Psi \circ \Phi(x_i) \right|\right|^2$$

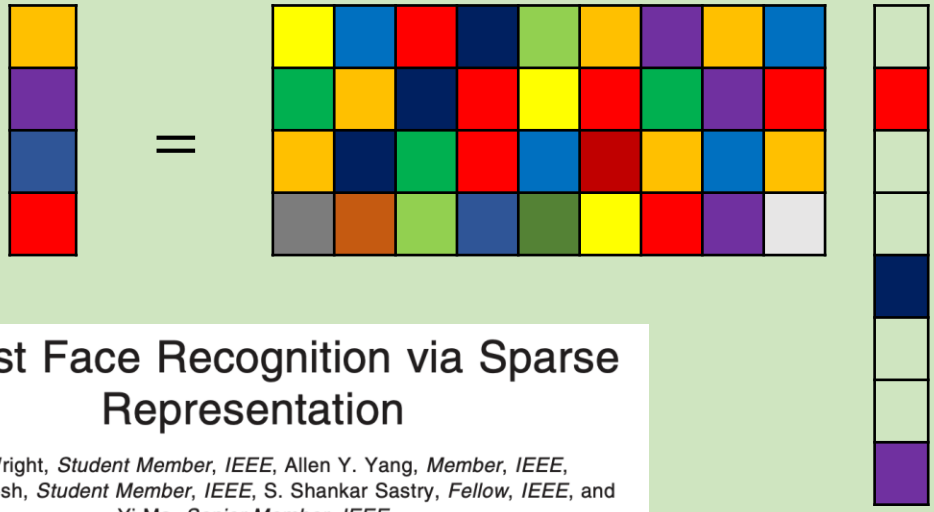
$\Psi = A \in \mathbb{R}^{d \times k}, k \gg d$, overcomplete dictionary

Φ – sparse code corresponding to x

Sparse Representation Learning

- Learn a "dictionary" such that each vector has a sparse representation in the dictionary

- Use the sparse coefficients for classification



Robust Face Recognition via Sparse Representation

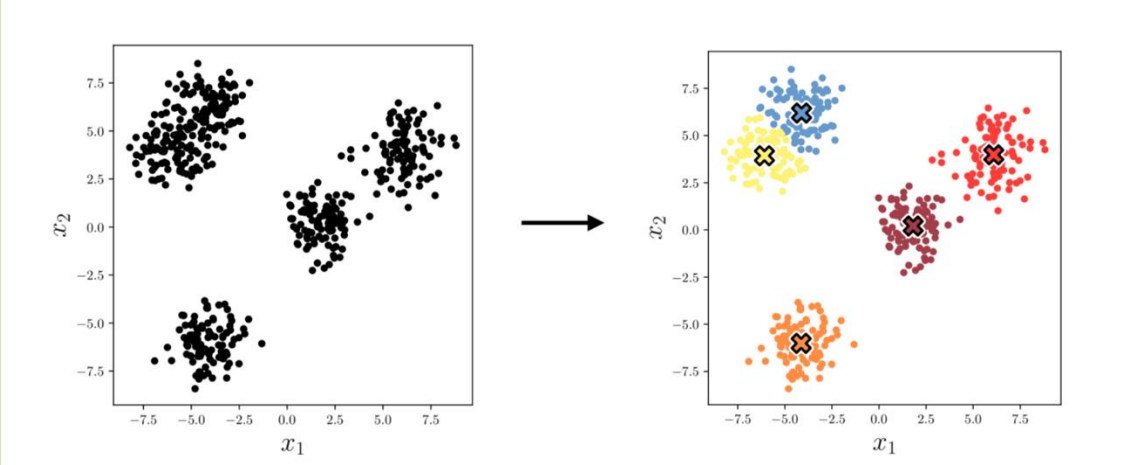
John Wright, *Student Member, IEEE*, Allen Y. Yang, *Member, IEEE*, Arvind Ganesh, *Student Member, IEEE*, S. Shankar Sastry, *Fellow, IEEE*, and Yi Ma, *Senior Member, IEEE*

K-means clustering

$$\min_{\Psi,\Phi} \frac{1}{n} \sum_{i=1}^{n} \left|\left|x_i - \Psi \circ \Phi(x_i)\right|\right|^2$$

Ψ – matrix of cluster means

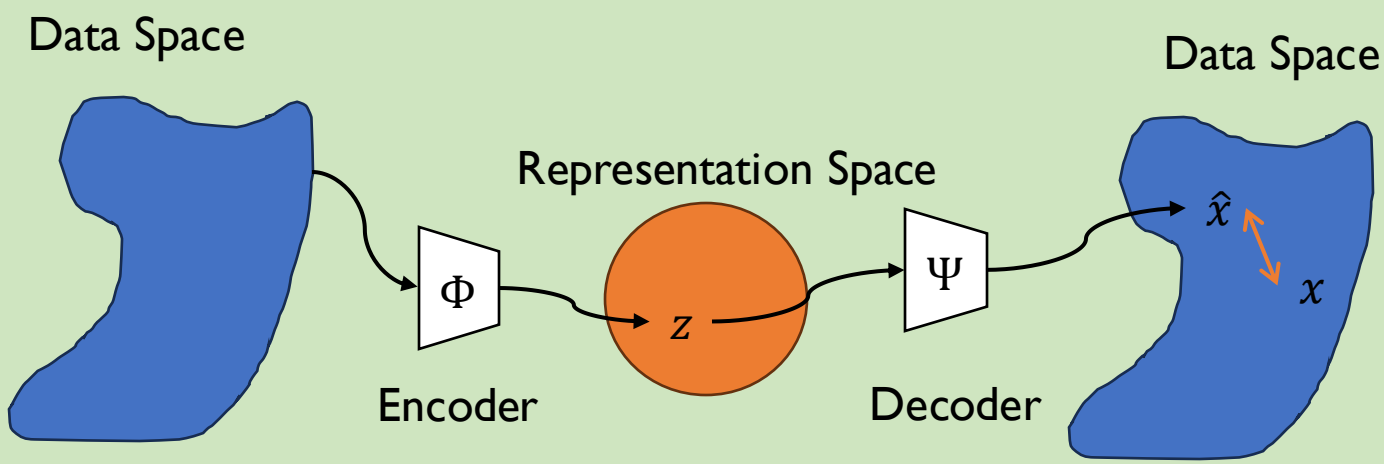
Φ – one-hot vector indicating cluster membership



Nonlinear Representation Learning

Reconstruction based Representation learning

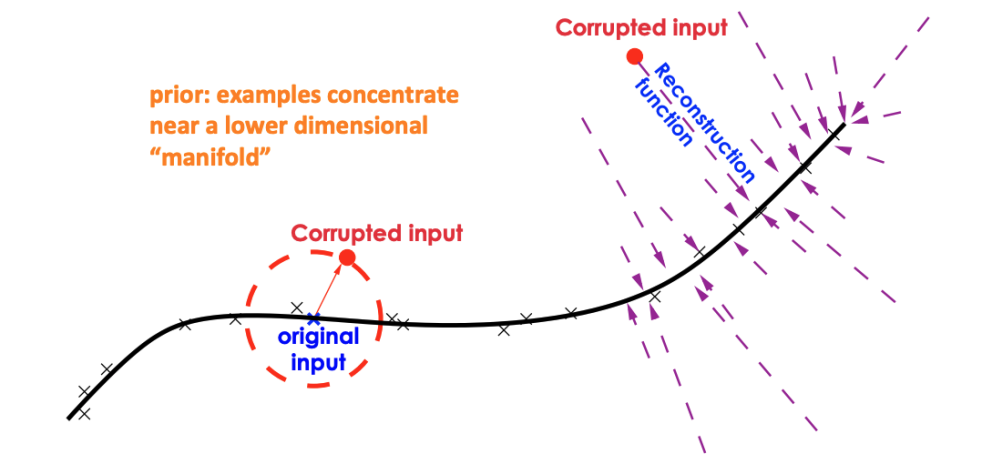
$$\min_{\Psi,\Phi} \frac{1}{n} \sum_{i=1}^{n} \left|\left| x_i - \Psi \circ \Phi(x_i) \right|\right|^2$$



Autoencoders

- $\Psi: \mathbb{R}^k \to \mathbb{R}^d, \Phi: \mathbb{R}^d \to \mathbb{R}^k$ are neural networks
- Typically, $k \leq d$, which means the encoder tries to "compress" the input into a smaller dimensional space
- If $k \leq d$ and the encoder/decoder are linear – equivalent to PCA
- If $k \gg d$, this is a "Sparse Autoencoder" and a sparsity penalty is typically applied on the hidden layer
- ReLU autoencoders perform sparse coding [Rangamani et. al. 2018]

Denoising Autoencoders



- Draw samples from the dataset
- Corrupt with some noise
- Reconstruct clean input with autoencoder

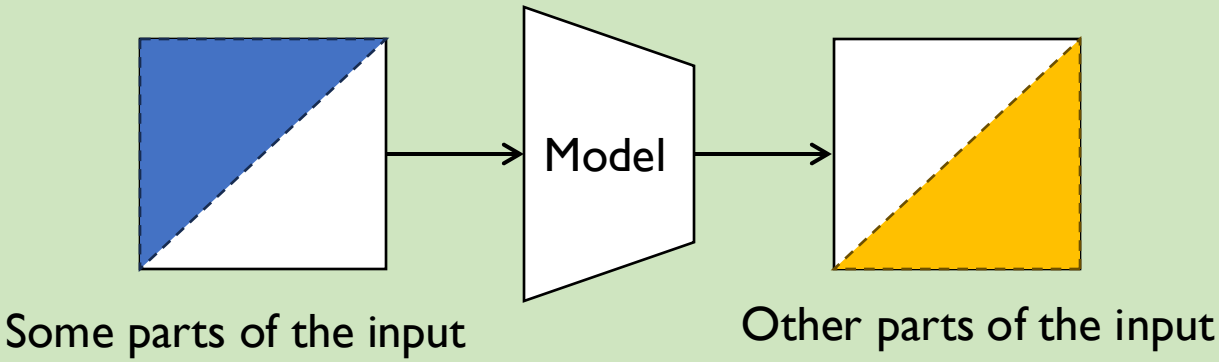
$$\min_{\Psi,\Phi} \frac{1}{n} \sum_{i=1}^{n} \mathbb{E}_{q(\tilde{x}|x_i)} \left[\left|\left| x_i - \Psi \circ \Phi(\tilde{x}) \right|\right|^2 \right]$$

Self-Supervised Learning

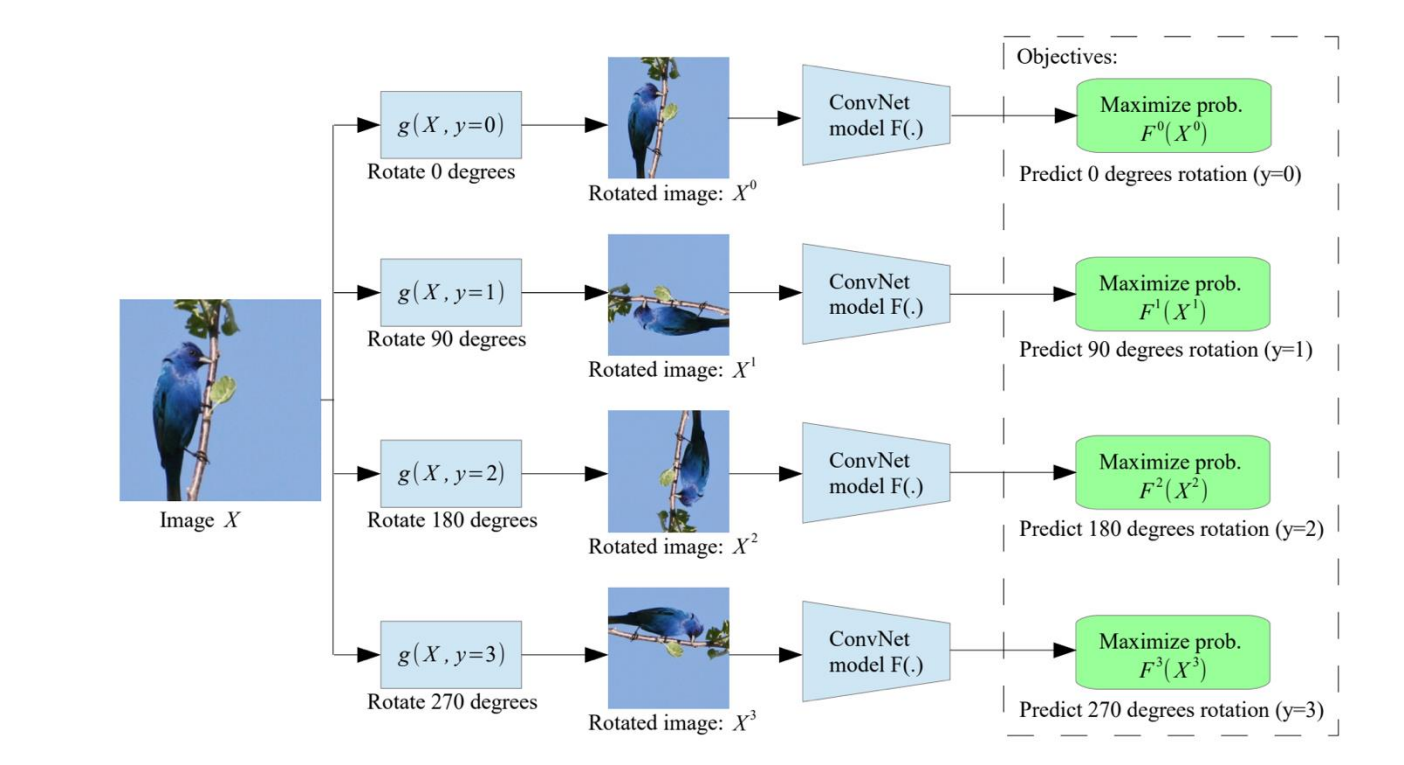
Self-supervised learning

- We have access to lots of unlabeled data, and not much labeled data

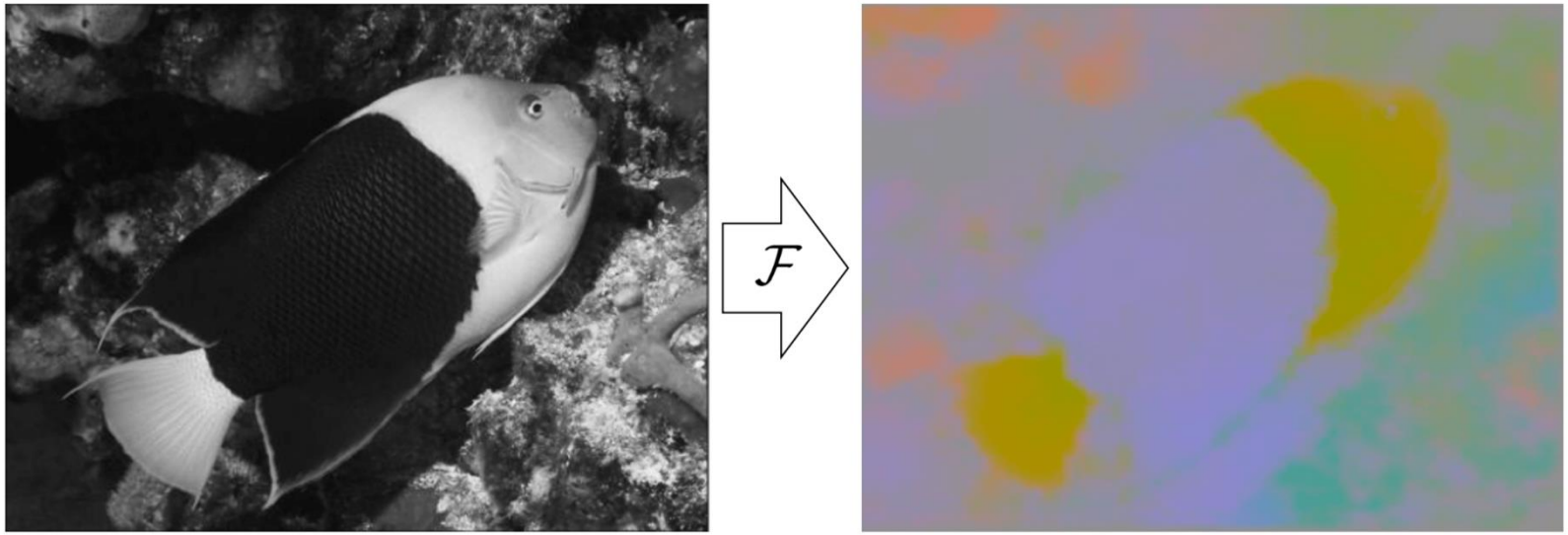
- Construct an "auxiliary task" from the unlabeled data
  - Gathering labels for this auxiliary task is easy/trivial



Auxiliary Tasks – Rotations [Gidaris et al 2018]



Auxiliary Tasks – Colorization

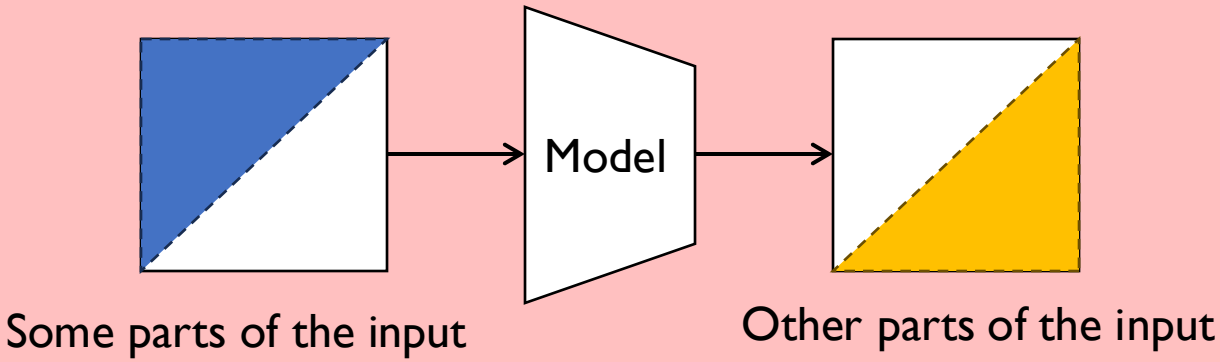


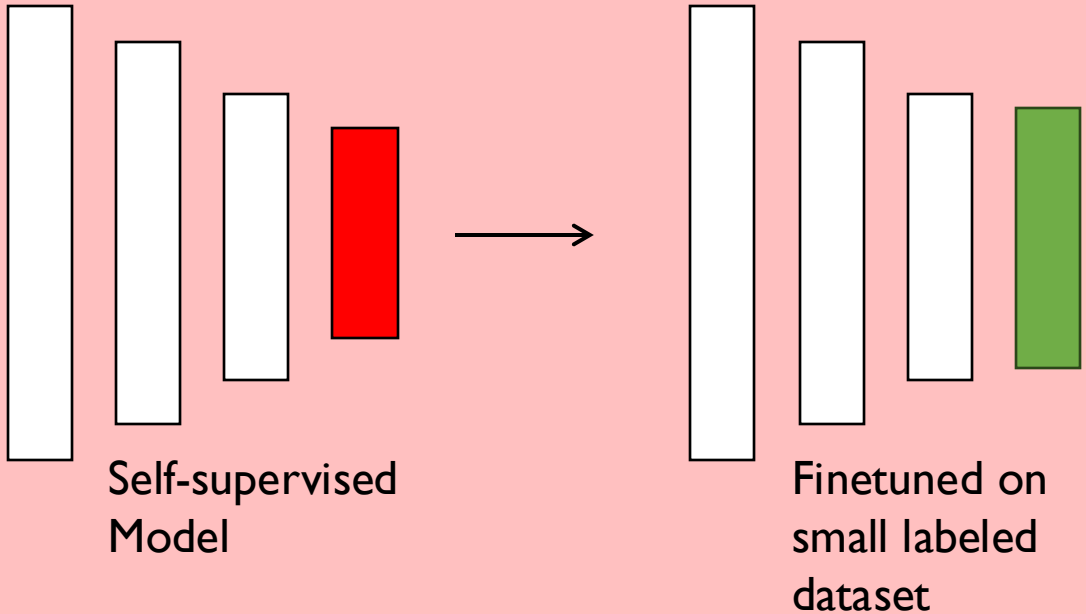
Grayscale image: L channel
$\mathbf{X} \in \mathbb{R}^{H \times W \times 1}$

Color information: ab channels
$\widehat{\mathbf{Y}} \in \mathbb{R}^{H \times W \times 2}$

[Zhang, Isola, Efros, ECCV 2016]

Using representations downstream



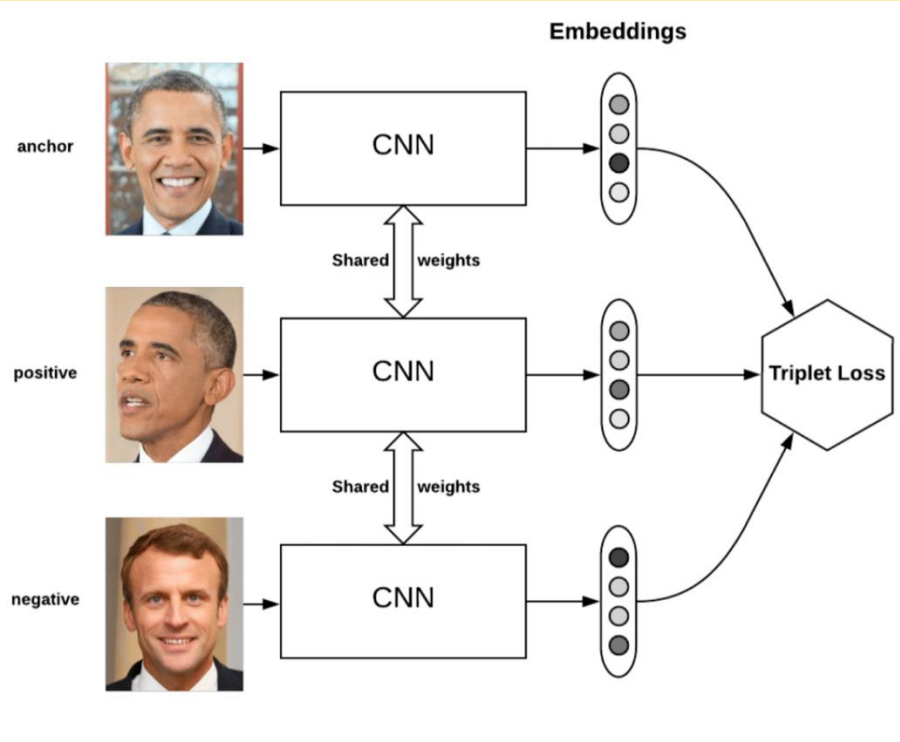


Contrastive Representation Learning

Contrastive Losses

- Distance of dissimilar points >> Distance of similar points

- Triplet loss – uses anchor, positive ex, negative ex
- $L = \sum_{x \in S} \max\left(0, \left|\left|f(x) - f(x^+)\right|\right|^2 - \left|\left|f(x) - f(x^-)\right|\right|^2 + \epsilon\right)$



Contrastive Loss

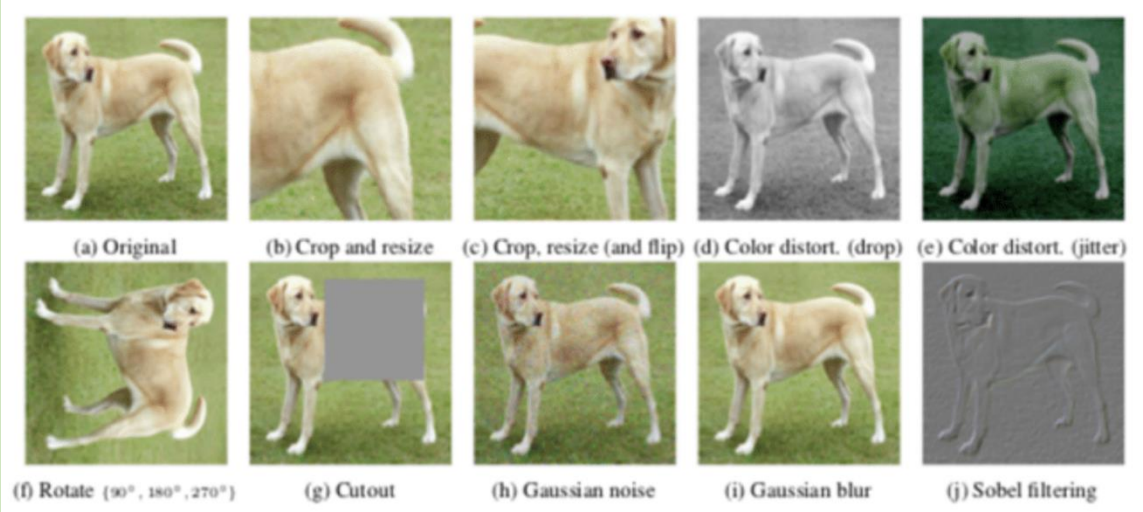
- Cross-entropy loss for identifying similar pairs

$$L = \mathbb{E}_{(x,x^+)\sim p_{pos},\{x_i^-\}\sim p_{data}}\left[-\log\frac{e^{\frac{f(x)^T f(x^+)}{t}}}{e^{\frac{f(x)^T f(x^+)}{t}}+\sum_{i=1}^{N} e^{\frac{f(x)^T f(x_i^-)}{t}}}\right]$$

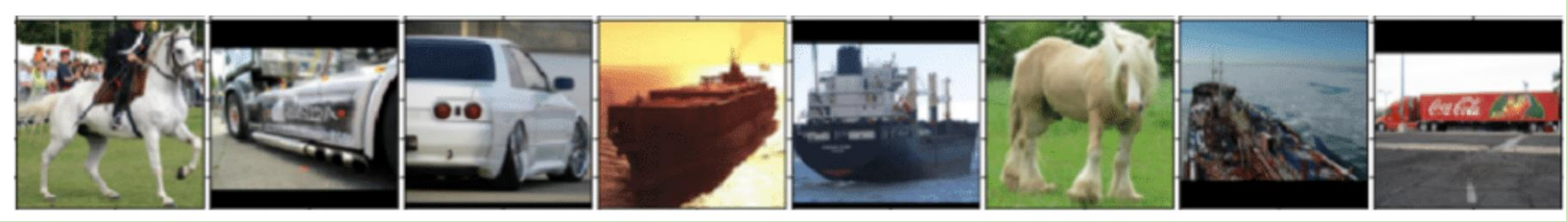
- Noise Contrastive Estimation (Gutmann & Hyvärinen 2010), InfoNCE (van den Oord et al 2018)

Contrastive Loss

- Positive examples – augmentations of the same sample



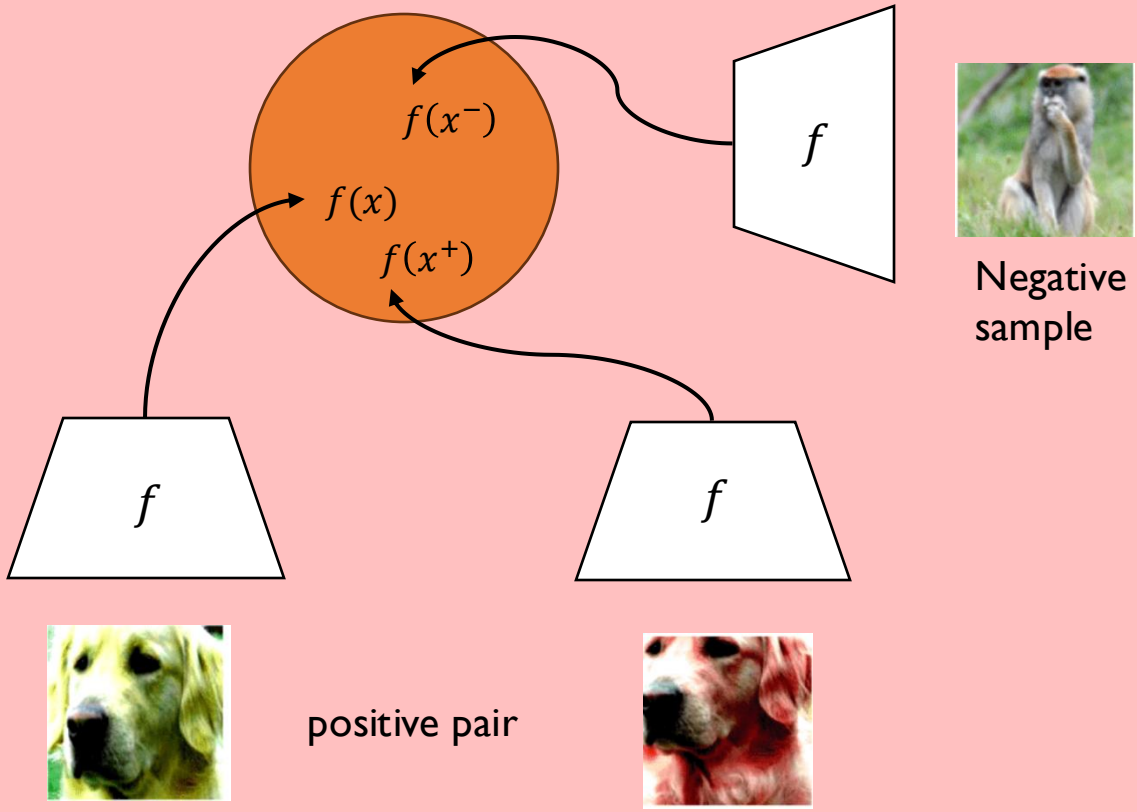
- Negative examples – other samples from the dataset



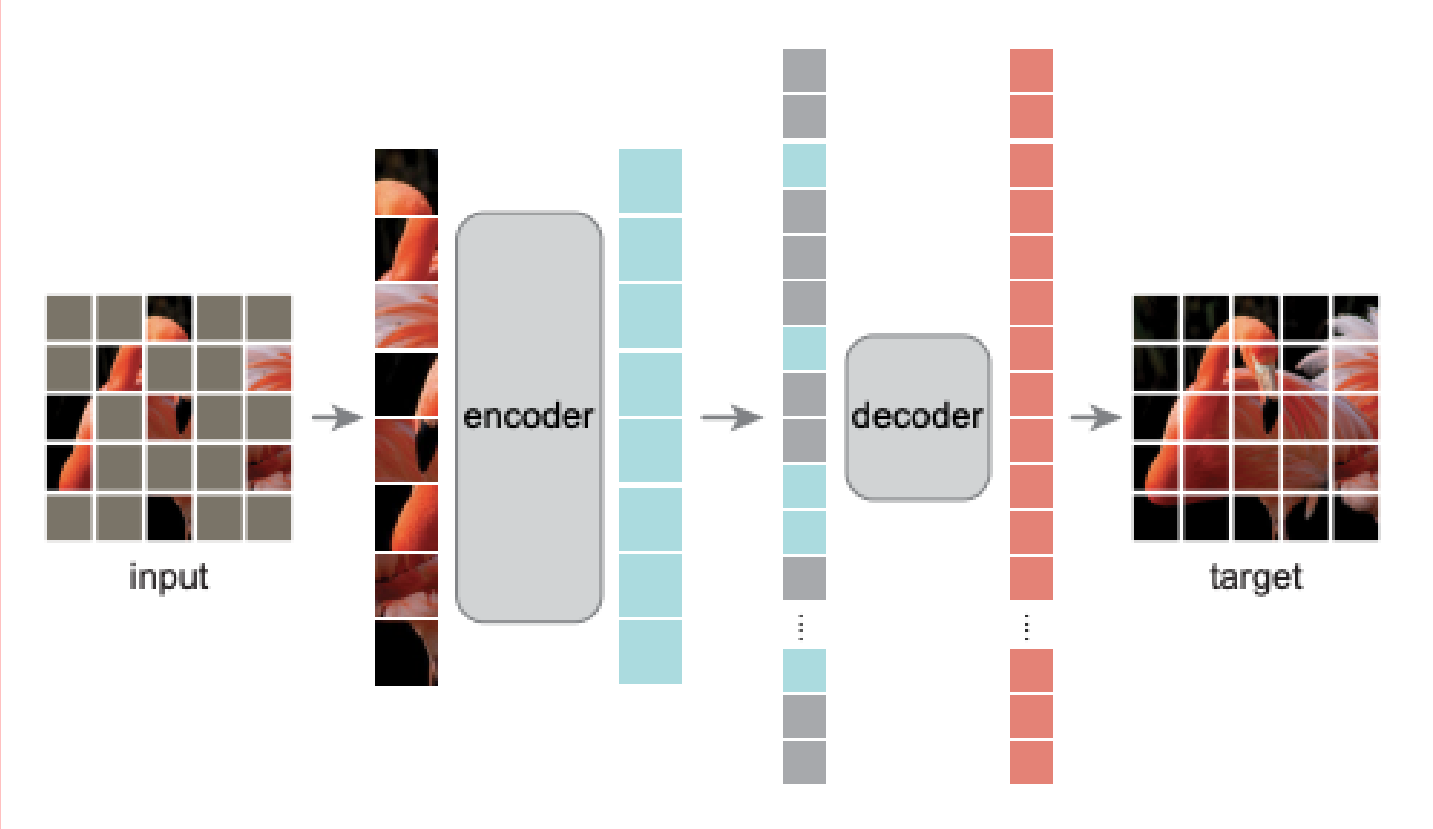
General Principle

- Cross-channel representation learning [CMC, Tian, Krishnan, Isola 2020]
- Video representation learning - [van den Oord, Li, Vinyals 2018]
- Language-Vision representation learning [Radford, Kim, et. al. 2021]

Contrastive Learning



MAE



He, K., Chen, X., Xie, S., Li, Y., Dollár, P., & Girshick, R. (2021). Mae: Masked autoencoders are scalable vision learners. *arXiv preprint arXiv:2111.06377*, 7.

MAE – higher masking ratios

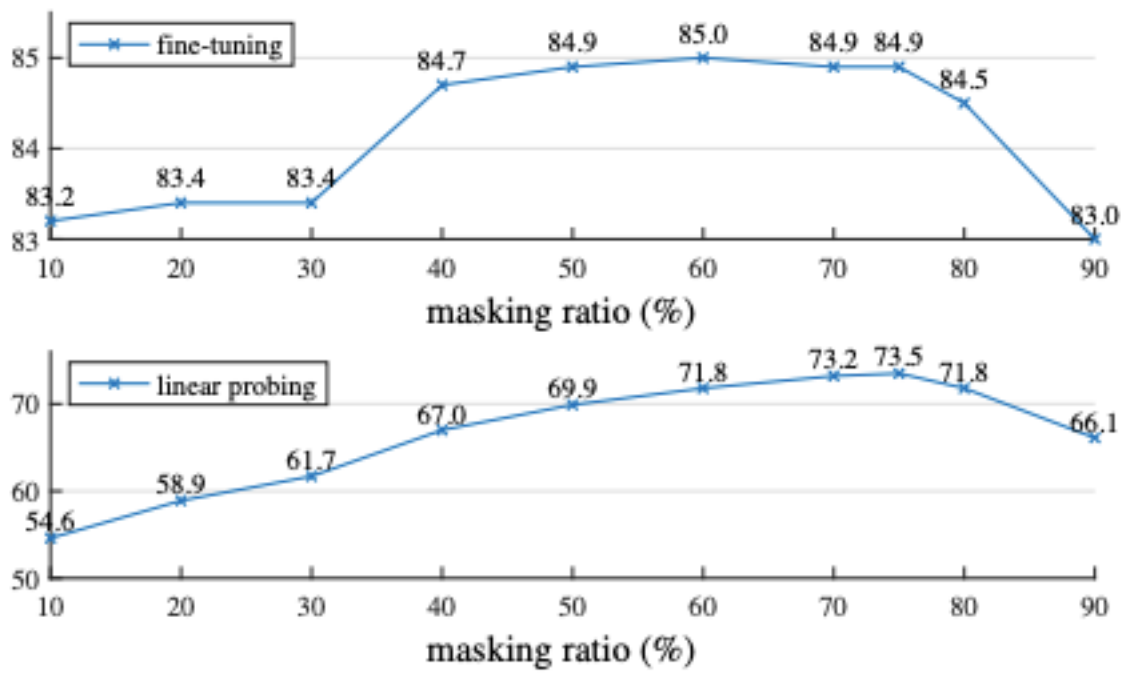


Figure 5. **Masking ratio**. A high masking ratio (75%) works well for both fine-tuning (top) and linear probing (bottom). The y-axes are ImageNet-1K validation accuracy (%) in all plots in this paper.

Not enough masking makes the reconstruction task too simple

MAE

blocks	ft	lin
1	84.8	65.5
2	**84.9**	70.0
4	**84.9**	71.9
8	**84.9**	**73.5**
12	84.4	73.3

(a) **Decoder depth**. A deep decoder can improve linear probing accuracy.

dim	ft	lin
128	**84.9**	69.1
256	84.8	71.3
512	**84.9**	**73.5**
768	84.4	73.1
1024	84.3	73.1

(b) **Decoder width**. The decoder can be narrower than the encoder (1024-d).

case	ft	lin	FLOPs
encoder w/ [M]	84.2	59.6	3.3×
encoder w/o [M]	**84.9**	**73.5**	**1**×

(c) **Mask token**. An encoder without mask tokens is more accurate and faster (Table 2).

case	ft	lin
pixel (w/o norm)	84.9	73.5
pixel (w/ norm)	**85.4**	**73.9**
PCA	84.6	72.3
dVAE token	85.3	71.6

(d) **Reconstruction target**. Pixels as reconstruction targets are effective.

case	ft	lin
none	84.0	65.7
crop, fixed size	84.7	73.1
crop, rand size	**84.9**	**73.5**
crop + color jit	84.3	71.9

(e) **Data augmentation**. Our MAE works with minimal or no augmentation.

case	ratio	ft	lin
random	75	**84.9**	**73.5**
block	50	83.9	72.3
block	75	82.8	63.9
grid	75	84.0	66.0

(f) **Mask sampling**. Random sampling works the best. See Figure 6 for visualizations.

Table 1. **MAE ablation experiments** with ViT-L/16 on ImageNet-1K. We report fine-tuning (ft) and linear probing (lin) accuracy (%). If not specified, the default is: the decoder has depth 8 and width 512, the reconstruction target is unnormalized pixels, the data augmentation is random resized cropping, the masking ratio is 75%, and the pre-training length is 800 epochs. Default settings are marked in gray.

BYOL

Compare and contrast with SimCLR

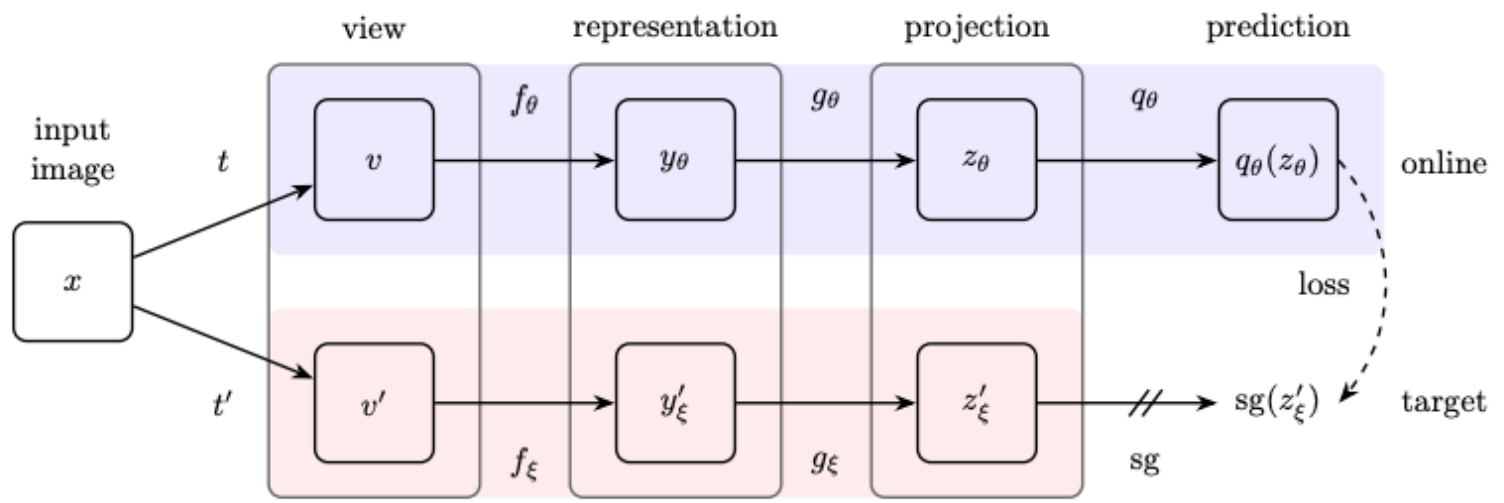
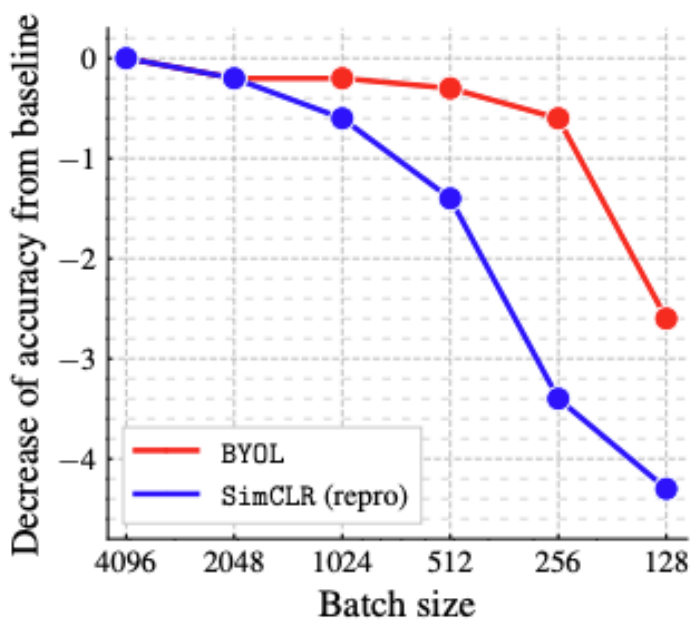


Figure 2: BYOL's architecture. BYOL minimizes a similarity loss between $q_\theta(z_\theta)$ and $\mathrm{sg}(z'_\xi)$, where θ are the trained weights, ξ are an exponential moving average of θ and sg means stop-gradient. At the end of training, everything but f_θ is discarded, and y_θ is used as the image representation.

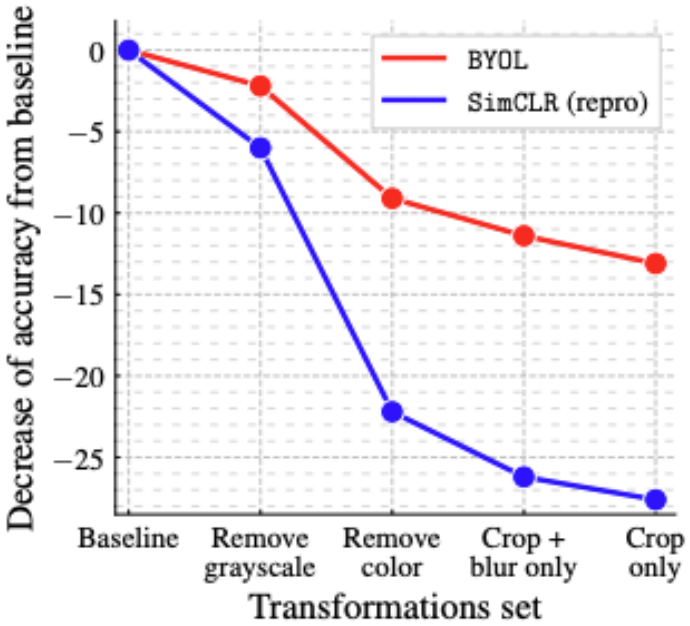
BYOL

https://imbue.com/research/2020-08-24-understanding-self-supervised-contrastive-learning/

BYOL



(a) Impact of batch size



(b) Impact of progressively removing transformations

Figure 3: Decrease in top-1 accuracy (in % points) of BYOL and our own reproduction of SimCLR at 300 epochs, under linear evaluation on ImageNet.

Multimodal Models

- CLIP - https://arxiv.org/pdf/2103.00020
- M3AE - https://arxiv.org/pdf/2205.14204
- Where does the robustness of CLIP come from?
  - https://arxiv.org/pdf/2205.01397
  - https://www.arxiv.org/pdf/2510.11835